

Selected Hyperparameter Settings

Model	Horizon (h)	No. of Features	Best Validation RMSE	Best Parameters
CatBoost	1	37	71.1867	iterations = 2590; learning_rate = 0.123665; depth = 5; l2_leaf_reg = 1.361687; subsample = 0.634694
XGBoost	1	37	69.8261	n_estimators = 2787; learning_rate = 0.019535; max_depth = 8; subsample = 0.774071; colsample_bytree = 0.991974; reg_lambda = 2.143587
Random Forest	1	37	75.0664	n_estimators = 772; max_depth =

				25; min_samples_split = 2; min_samples_leaf = 1; max_features = 0.8
LSTM	1	37	77.3976	units1 = 74; units2 = 95; dropout = 0.113063; lr = 0.003400; batch_size = 32
ARIMA	1	37	1559.0764	order = (4, 1, 2)
CatBoost	6	36	215.3778	iterations = 1946; learning_rate = 0.035250; depth = 10; l2_leaf_reg = 5.443974; subsample = 0.838889
XGBoost	6	36	216.7362	n_estimators = 2724; learning_rate = 0.010089;

				max_depth = 10; subsample = 0.797964; colsample_bytree = 0.802900; reg_lambda = 4.800670
Random Forest	6	36	217.4922	n_estimators = 743; max_depth = 28; min_samples_split = 4; min_samples_leaf = 2; max_features = 0.5
LSTM	6	36	211.1865	units1 = 72; units2 = 126; dropout = 0.263467; lr = 0.001831; batch_size = 32
ARIMA	6	36	2281.1974	order = (5, 1, 2)
CatBoost	12	35	239.6215	iterations = 1160; learning_rate =

				0.012691; depth = 10; l2_leaf_reg = 1.187233; subsample = 0.931727
XGBoost	12	35	242.4469	n_estimators = 507; learning_rate = 0.061114; max_depth = 8; subsample = 0.627137; colsample_bytree = 0.716535; reg_lambda = 9.095323
Random Forest	12	35	244.3143	n_estimators = 269; max_depth = 27; min_samples_split = 10; min_samples_leaf = 7; max_features = log2

LSTM	12	35	247.1657	units1 = 124; units2 = 125; dropout = 0.388816; lr = 0.002422; batch_size = 32
ARIMA	12	35	897.3462	order = (5, 1, 5)
CatBoost	24	34	304.5487	iterations = 1022; learning_rate = 0.067223; depth = 10; l2_leaf_reg = 9.963748; subsample = 0.758847
XGBoost	24	34	308.5548	n_estimators = 1183; learning_rate = 0.015140; max_depth = 6; subsample = 0.609423; colsample_bytree = 0.985540;

				reg_lambda = 5.455652
Random Forest	24	34	310.0743	n_estimators = 403; max_depth = 28; min_samples_split = 12; min_samples_leaf = 7; max_features = sqrt
LSTM	24	34	308.0061	units1 = 44; units2 = 110; dropout = 0.442001; lr = 0.004505; batch_size = 32
ARIMA	24	34	1753.6863	order = (5, 1, 1)